

Distributed Search Migration Lab – Lab Guide

IMPORTANT: The **approximate build time for this lab is 10 minutes**. Please be patient. If the lab does not build after 15 minutes, try refreshing the page and rebuilding it again. If it does not build on the second attempt, contact ellearn@splunk.com.

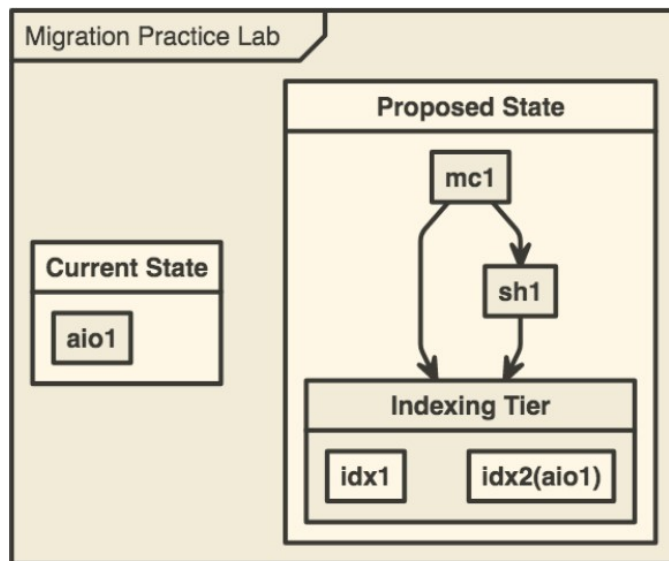
Lab Overview

Maximum Estimated Duration – 4 Hours

In this lab, your aim is to emulate the process of expanding a Splunk infrastructure from a single all-in-one node to a distributed (multi-indexer, plus search head) configuration.

The various Splunk hosts used within this exercise have the following roles:

- **aio1** – This represents a host that used to be a single "all-in-one" Splunk host which until now has done all the work of the Splunk infrastructure – indexing and searching.
- **idx1** – This is a new instance to be initialized as a primary indexer within this environment. Per the narrative above, the goal is to expand the infrastructure to provide more capacity. This indexer will ingest data but will need to do so in a fashion consistent with the "singleton" node described above.
- **sh1** – This node is a new instance, which is expected to fit the role of a "search head" (hence "sh") in a new distributed environment. It will be connected to the indexers to search the data contained there. It will also serve in the role of a Deployment Server (DS) to ensure consistent configuration across the other nodes.
- **mc1** - The Management Console Node you will use to monitor your environment and run the health checks on.



Deploy and Install Splunk on the New Instances – idx1, sh1, and mc1

In this section you will use the same steps to configure your infrastructure as in the first practice lab, Services Consultant Architecture Practice Lab 1.

Perform the following on each host:

1. Create a splunk user
2. Unarchive Splunk Enterprise package in /opt.
3. Install Splunk as Root
4. Change the default Admin password (Use the password provided in the Server List)
5. Run Splunk as a non-root user – splunk
6. Enable Splunk to start on boot as the splunk user

Prepare Deployment Server Content

In this exercise, you'll use the base configs to seed the content of the deployment server.

Retrieve the latest contents of the base configs from

<https://drive.google.com/drive/folders/107qWrfsv17j5bLxc21ymTagjtHG0AobF>

You'll be customizing several of these to suit your deployment. For any subsequent changes, replace the "org" prefix with the lab identifier "cfrtl4", so that the apps are easily recognizable.



Prepare an app to manage the location of the deployment server itself:
`org_all_deploymentclient`.

You'll need to use the bare (public) IP address, don't count on DNS in this environment.

1. Tweak the following apps as needed to prepare your indexer's configuration:
 - **org_all_indexer_base** – Baseline settings for an indexer, such as input port, disabling Splunkweb, etc.
 - **org_all_indexes** – Index definitions
 - **org_full_license_server** – Location of the license master (Use SH1 for the license master - the Trial license will suffice for this lab).
 - **org_indexer_volume_indexes** – Defines the main filesystem path (volume) for use with the `org_all_indexes` app; can vary between indexers and search heads, hence a separate app.
2. Tweak the following apps as needed to prepare your search head's configuration:
 - **org_all_search_base** – Base functionality for a search head; this includes LDAP authentication (not used in this environment), custom roles (`authorize.conf`) and a choice to make the web interface use SSL (`web.conf`)
 - **org_all_forwarder_outputs** – List of target indexers that gets shipped to any forwarder; this way, all data makes it to the indexer pool, and there's only one place to edit when the indexer list changes.
 - **org_search_volume_indexes** – Satisfy the "volume:primary" definition in the indexes app, but with a path suitable for the search head (and likely no volume limit, since it won't index locally).

Placing your customized versions of these apps within the `$SPLUNK_HOME/etc/deploymentapps` folder of your DS instance is sufficient to kick start the process of selng up "Forwarder Management" to distribute these apps to the clients.

To copy directories to your instances, use 'scp'. This sample command will copy the contents of the "org_all_indexes" directory (assumed to be a child of the present directory) to the apex host's "/tmp" directory:

```
scp -r org_all_indexes sccStudent@<NAME>-idx1:/tmp
```

More deployment content will be retrieved from the "aio" Splunk instance you started with during the following module.

[Configure the New Indexer and the MC as deployment clients](#)

In this section of the lab, you'll begin to build out a distributed search architecture by setting up a new indexer to provide additional indexing capacity.

1. Copy your customized `org_all_deploymentclient` application to the `$SPLUNK_HOME/etc/apps` folder of this new indexer. This is required to "touch the host once" to establish the deployment client / server relationship.
2. Connect to the node identified above as "idx1", creating an active session.
3. Restart Splunk for this change to take effect.
4. Repeat for mc1

Check Your Work

1. Log in to the GUI of the DS instance (ds).
2. Click on Settings -> Forwarder Management. Select the "Clients" tab.
3. Validate that the IP (or hostname) of the indexer and the mc you just set up is present in the list of clients that have phoned home.

You'll work to distribute apps to these hosts in a later portion of the lab.

Troubleshooting Suggestions

If the indexer does not appear as a deployment client in the "Forwarder Management" screen, or it looks like this screenshot, consider the following steps.

Ensure that you have copied apps into the `$SPLUNK_HOME/etc/deployment-apps` folder of the DS instance, then visit the Forwarder Management page again.

Double-check that the indexer has been restarted since installing the deployment client app. The following command should produce a line declaring the role of `deployment_client`: `grep ServerRoles $SPLUNK_HOME/var/log/splunk/splunkd.log`.

Use `btool` to verify that the deployment server URI (in the conf) is correct. The `targetUri` `selng` should not contain a protocol; just an IP address and port number. This port number is for the management port, not the Splunkweb port.

Reclaim Settings from an Existing Node

In this portion of the lab you will harvest any knowledge objects and/or configuration settings from the standalone node (aio1) during the Onboarding lab. The purpose is to make these changes available to all indexers (for configuration and index-time parsing rules) as well as to the search heads (search-time props entries).

1. Establish an interactive session on the original standalone node (singleton).
2. Edit the `$SPLUNK_HOME/etc/system/local/server.conf` selng in [general] for `serverName`; change this to `<your_name>_idx2`, e.g. `"cfrtl4_1234_idx2"`.
3. Use `btool` in debug mode to identify where the settings for the "onboarding" index was written by the GUI.
4. Collect these settings and add them to your customized `org_all_indexes` app on the DS.
5. Update the `homePath` and `coldPath` as needed to reflect the volume specification from that template app.
6. Again, leveraging `btool`, locate the `props.conf` and `transforms.conf` (if any) for the sourcetypes listed here: `volume-iops`, `edifecs`, `sshd_syslog` and `iis_onboarding`.

For each of these:

7. Create a customized version of the `"org_all_<app>_props"` app for the appropriate sourcetype by cloning (recursive copy will do) the `"org_APP_TEMPLATE"` app.
8. Use the following naming convention for these four apps: `cfrtl4_sourcetype_*_props`. For example, `cfrtl4_sourcetype_edifecs_props`.

This template contains enough information to share the objects in the app globally (metadata/local.meta), and also to satisfy the deployment server (local/app.conf).

9. Place the `props.conf` and `transforms.conf` in this app's local folder.

There is no need to attempt to separate any index-time rules from search-time rules; Splunk will ignore what the instance doesn't need.

Make the Single Instance a Deployment Client

In this lab exercise, the time has come to convert the single instance system you began with into a deployment client. This will allow the configuration files of this instance to be centrally managed.

1. Stop Splunk.
2. Remove (or rename, to something not ending in `.conf`) the `props` and `transforms` files you identified in the previous section.
3. Copy your "all deployment client" app to the `$SPLUNK_HOME/etc/apps` directory.

4. Restart Splunk.

NOTE: There will still be configuration files created or edited while the aio host was a standalone host. After the next lab (using the DS to ensure that the configurations are uniform), you can clean out the old files with no harmful effects.

Use the DS to Ensure Uniform Configuration

In this section you will utilize the Deployment Server (also known as Forwarder Management) to ensure that the two indexers in your environment have a consistent configuration, and you will also send some basic configurations to your new search head.

1. Log on to the Deployment Server's GUI. Navigate to Settings > Forwarder Management.

By this point, you should see three clients connecting; the SH and two indexers.

2. Create a new serverclass (collection of hosts) by going to the Server Classes tab and clicking the link to "create one".
3. First create a class matching all hosts - name it `all_hosts`; this will be used for keeping the `org_all_deploymentclient` itself up to date. Add only this app to the class. For hosts, enter a `*` (wildcard) on the "Include" section of the list. You can click "Preview" to see which hosts would match this class. Click "Save" when you are done.
4. Next create a class for the indexers, call it `all_indexers`. In the next screen, add the following apps to the class: `org_indexer_volume_indexes` `org_all_indexes` `org_all_indexer_base` `org_sourcetype_*_props` (for each of the apps you created in section four above). `org_full_license_server`
5. Add the indexers (as clients) to this class, one by one and "Save".
6. Finally, create a class for the search head and call it `all_search_heads`. Similar to the procedure above, you will add the following apps to this class: `org_all_search_base` `org_all_forwarder_outputs` `org_all_indexes` `org_search_volume_indexes` `org_sourcetype_*_props`
7. The Monitoring Console is the appropriate client for this class, so add it in the client list and then "Save".

The search head also requires these apps but as it is the deployment server it cannot deploy to itself.

8. Copy the apps to the `etc/apps` folder of your search head.
9. Restart the instance to apply the configuration.

10. Log in to one or two of the hosts and observe that their `$SPLUNK_HOME/etc/apps` folders get populated with new content.

Certain changes to the configuration of Splunk can be installed smoothly via the DS without restarting Splunk. By contrast, other changes do require a restart. Included in the latter list are `conf`, `outputs.conf` and `indexes.conf` changes. Because nearly all of these changes were made to one host or another, you will need to restart Splunk on your search head, single and indexer2 instances, after validating that their application content has arrived from the DS.

Set Up Distributed Search on the Search Head

In this exercise, you will complete the distributed search environment by establishing the search peer relationship between the search head and the indexers.

1. Log in to the SH GUI.
2. Click Settings -> Distributed Search
3. In the line that reads "Search Peers ... Add new", click "Add new".

The address of an indexing peer is given as `protocol://host_or_ip:port`; in this case we're using the default HTTPS on the management port of Splunk. That default port number is 8089. The remote username and password are for the indexer itself, not the instance you are logged in to.

4. After entering the username and password twice, click "Save".

The admin password for the existing node is the same as the lab password for `sccStudent`

NOTE: The credentials used (username and password) are used for key exchange and require admin privileges on the target indexer. These credentials are not stored; the certificate exchange, done once, establishes host-based trust.

5. Repeat for the second indexer.

Configure the Monitoring Console

Configure `idx1`, `aio1`, and `sh1` as search peers of `mc1` using the same steps in the previous section.

To monitor your configured environment, you will need to configure your monitoring console in distributed mode.

Further information on the Monitoring Console can be found here:

[h@p://docs.splunk.com/Documentation/Splunk/latest/DMC/DMCoverview](http://docs.splunk.com/Documentation/Splunk/latest/DMC/DMCoverview)

Instructions on configuring the monitoring Console in distributed mode can be found here:

<http://docs.splunk.com/Documentation/Splunk/latest/DMC/Configureindistributedmode>

Troubleshooting

If you are unable to add the other instances as search peers, consider the following:

The connection to the remote indexer to perform the certificate exchange uses as RESTful query. A newly installed Splunk instance will refuse any REST query until either its admin password has been changed, or a specific option is set to allow this kind of “login with default credentials”. This is a protection mechanism built into Splunk. Have you changed the admin password?

Check Your Work

Issue a search on the `_internal` index from the SH GUI, observe the number of values of the “`splunk_server`” field. How many are there? Why?

Validate the roles of each host.

Run a search on the search head: “`| rest /services/server/info`”. You should find three nodes.

The keys returned by this search command are returned lexically, with those starting with capital letters coming first. Le9 to right, validate the following:

host – Should reflect the base internal name of the instance

serverName – Reflects the value of the `server.conf` [general] `serverName` parameter; can be updated to make identifying the nodes easier.

server_roles – This multivalue field lists the self-identified roles of each node. From the SH, the search head node itself should identify as “`search_head`”. The indexers should identify as both “`indexer`” and “`search_peer`”. All of the nodes should identify as a “`deployment_client`”.

Troubleshooting

If a host is not reflecting the right roles, consider the steps below:

If your hosts don't reflect the right server roles, consider the means by which Splunk determines this.

The "indexer" role is set if a host is a full instance of Splunk but does not have any config listing remote servers. This means that the host would be indexing its data locally.

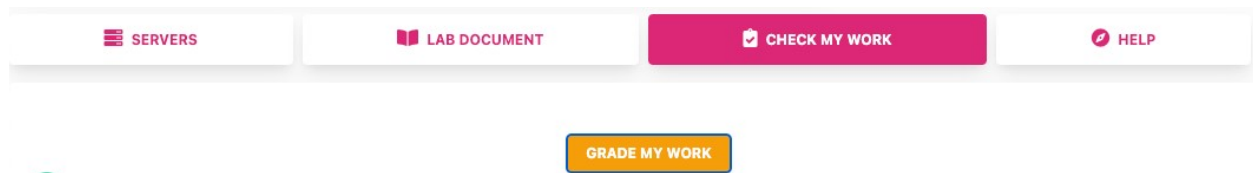
If a deploymentclient.conf file is found when Splunk starts, it initializes a body of code to act as a deployment client and sets the server role accordingly. This information will come in handy during the indexer clustering lab.

For the server side of the deployment server / deployment client functionality, the presence of serverclass.conf is what triggers the deployment server code line, and the "deployment_server" role.

Setting up distributed search peers (and searching from that host) makes it a "search_head"; the peers become "search_peer's.

Grading Your Work

Grading of the lab is done through the same interface where your student instances are displayed.



Grading of the labs is an intensive process involving many calls to the instances. This may take several minutes to complete. Please be patient. If the grading does not complete in 5 minutes, please reach out to ellearn@splunk.com for assistance.

For best results, make sure you complete all of the lab steps before grading the lab.

1. Click the Check My Work Tab.
2. Click the Grade My Work button.

After a few minutes, your results will be displayed.



You must pass all graded items to receive credit for the lab. Resolve any failed grading items and re-issue the grading check until you receive a green check mark for all graded items and a large green check mark at the bottom.



Congratulations, you have completed all lab work.

Please take a screenshot of the grading page showing all of the items green and save for your records. In the event the Learning Management System does not mark the lab complete you can send a request to education@splunk.com requesting it be manually marked complete. Make sure to include the screenshot you took showing the lab was passed.

Here is a sample screenshot of a passed lab.

GRADE MY WORK

-  Splunk is installed correctly in /opt on Search Head 1.
-  Splunk is installed correctly in /opt on the Monitoring Console.
-  Splunk is installed correctly in /opt on Indexer 1.
-  Splunk is installed correctly in /opt on Indexer 2.
-  Indexer Receiving Ports are configured on Indexer 1.
-  Indexer Receiving Ports are configured on Indexer 2.
-  MC Search Head Role Assigned Correctly.
-  Indexer Base Template App Installed is installed on Indexer 1.
-  Indexer Base Template App Installed is installed on Indexer 2.
-  Search Head Distributed Search Configured for Indexer 1.
-  Search Head Distributed Search Configured for Indexer 2.
-  Monitoring Console Distributed Search Configured for Search Head 1.
-  Monitoring Console Distributed Search Configured for Indexer 1.
-  Monitoring Console Distributed Search Configured for Indexer 2.



Congratulations, you have completed all lab work.